POINTS, IRREDUCIBLE CLOSED SUBSETS AND INTEGRAL CLOSED SUBSTACKS OF A DECENT STACK WITH QUASI-COMPACT INERTIA CORRESPOND

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that the space of a decent algebraic stack is sober. Invoked at the start of the counted proof (source lines 10722-10723). Gives the bijection between points and irreducible closed subsets.
- The construction of the reduced induced closed substack structure on a closed subset. Invoked inside the counted proof (source lines 10725-10728) as a cross-chapter pointer. Produces the closed substack with a prescribed underlying closed subset.
- The permanence of decency for a representable morphism. Invoked inside the counted proof (source lines 10731-10732). Makes the reduced closed substack decent.
- The base change description of the inertia of a closed substack. Invoked near the end of the counted proof (source lines 10733-10735). Gives quasi-compactness of the inertia of the substack.

Source reference. Original lemma lemma-monomorphism-cartesian-square-inertia: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1480–1480. Navigation pointer only: the counted text refers to the Stacks item lemma-monomorphism-cartesian-square-inertia, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-representable-decent: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10425–10425.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

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Source reference. Original proposition proposition-decent-sober: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10575–10575. Navigation pointer only: the counted text refers to the Stacks item proposition-decent-sober, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference stacks-properties-definition-reduced-induced-stack: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6968–6968. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-definition-reduced-induced-stack, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be a decent algebraic stack such that $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is quasi-compact. There are canonical bijections between the following sets:*

(1) *the set of points of $\mathcal{X}$, i.e., $|\mathcal{X}|$,*
(2) *the set of irreducible closed subsets of $|\mathcal{X}|$,*
(3) *the set of integral closed substacks of $\mathcal{X}$.*

The bijection from (1) to (2) sends x to $\overline{\{x\}}$. The bijection from (3) to (2) sends $\mathcal{Z}$ to $|\mathcal{Z}|$.

Proof. Our map defines a bijection between (1) and (2) as $|\mathcal{X}|$ is sober by Proposition proposition-decent-sober. Given $T \subset |\mathcal{X}|$ closed and irreducible, there is a unique reduced closed substack $\mathcal{Z} \subset \mathcal{X}$ such that $|\mathcal{Z}| = T$, namely, $\mathcal{Z}$ is the reduced induced subspace structure on T, see Properties of Stacks, Definition stacks-properties-definition-reduced-induced-stack. Then $\mathcal{Z}$ is an integral algebraic stack because it is decent (Lemma lemma-representable-decent), the morphism $\mathcal{I}_\mathcal{Z} \to \mathcal{Z}$ is quasi-compact (as the base change of $\mathcal{I}_\mathcal{X} \to \mathcal{X}$, see Lemma lemma-monomorphism-cartesian-square-inertia), $\mathcal{Z}$ is reduced, and $|\mathcal{Z}|$ is irreducible. □